

Exact generating-function analysis of a two-class priority M/M/1 queue

with jockeying and abandonment

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Thesis de

§1 • MOTIVATION

Waiting lists that change while you wait

In elderly-care admission, medical severity sets a customer's **priority class**. But waiting is not passive: deteriorating health can push a waiting customer to **change class** or to **leave the system** — for alternative care, or through death.

priority

non-priority

jockeying

abandonment

How do jockeying and abandonment interact within a priority discipline — and how does that interaction affect our ability to solve the queue in closed form?

Three streams — and the gap between them

Queues with abandonment Erlang-A (Garnett et al., 2002); Kapodistria (2011) — exact priority-plus-reneging stays multi-server and waiting-time based. the special-function toolkit	Exact priority queues Cohen (1956); Zuk & Kirszenblat (2023) — exact, but mechanism-free: nobody moves, nobody leaves. the methodological ancestors	Jockeying & class change de Waal (1992) treats it approximately; Hu, Chan & Dong (2022) study it asymptotically, multi-server. the applied motivation
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The gap: an **exact, single-server** treatment admitting both mechanisms in one bivariate-PGF framework.



OUTLINE

Five parts

I	The toolbox	$M/M/1 \cdot M/M/1+M$ · generating functions · w derivatives come
II	Model X and its fundamental equation	the system · balance equations · the fundam equation · the empty-state the
III	The solved models	one theorem per model: $A \cdot B_2 \cdot C_2 \cdot B^H \cdot C^H$ – why B stays
IV	The $\tilde{S}$ experiment	an alternative state space, and a docum negative r
V	Results	CTMC validation · jockeying vs. abandonr across load · convergence to the bas



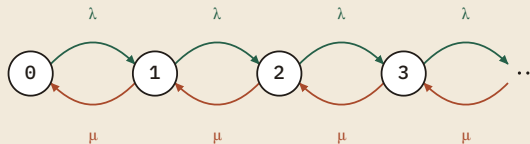
The toolbox

Two classical queues, one transform, and the mechanic that turns waiting-room behaviour into derivatives.

§3.1 • PRELIMINARIES

The M/M/1 queue: one chain, three formulas

BIRTH-DEATH CHAIN — ARRIVALS λ UP, DEPARTURES μ DOWN



DETAILED BALANCE $\Rightarrow$ GEOMETRIC LAW

$$\lambda \pi_n = \mu \pi_{n+1} \implies \pi_n = (1 - \rho) \rho^n, \quad \rho = \frac{\lambda}{\mu} < 1$$

MEAN QUEUE AND SOJOURN (LITTLE)

$$\mathbb{E}[L] = \frac{\rho}{1 - \rho}, \quad \mathbb{E}[W] = \frac{1/\mu}{1 - \rho}$$

BUSY PERIOD — REMEMBER THIS ONE

$$\widehat{BP}(s) = \frac{(\mu + \lambda + s) - \sqrt{(\mu + \lambda + s)^2 - 4\mu\lambda}}{2\lambda}$$

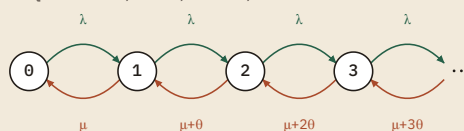
$$\mathbb{E}[BP] = \frac{1}{\mu(1 - \rho)}$$

The busy period returns later as the **effective service time** a class-2 customer experiences: the server must first drain the class-1 queue.

§3.3 • PRELIMINARIES

M/M/1+M: impatience brings in Kummer

IMPATIENCE CHAIN — DEPARTURES GROW WITH THE QUEUE: $\mu, \mu+\theta, \mu+2\theta, \dots$



WAITING CUSTOMERS ABANDON AT RATE θ EACH

$$\lambda \pi_n = (\mu + n\theta) \pi_{n+1} \implies \pi_n = \pi_0 \frac{(\lambda/\theta)^n}{(\mu/\theta)^{(n)}}$$

WHAT WE SOLVE FOR — IDLE PROBABILITY & QUEUE-LENGTH PGF

$$\pi_0 = \left[{}_1F_1\left(1; \frac{\mu}{\theta}; \frac{\lambda}{\theta}\right) \right]^{-1}, \quad \Pi(z) = \frac{{}_1F_1\left(1; \frac{\mu}{\theta}; \frac{\lambda}{\theta}z\right)}{{}_1F_1\left(1; \frac{\mu}{\theta}; \frac{\lambda}{\theta}\right)}$$

Stable for **every** load — the departure rate $\mu+n\theta$ grows without bound, so no $\rho < 1$ is needed.

THE KUMMER FUNCTION ${}_1F_1(A; B; Z)$

The confluent hypergeometric function — the entire solution of Kummer's equation $z u'' + (b-z)u' - a u = 0$ that is regular at $z = 0$.

POWER SERIES $\cdot (A)_n$ THE RISING FACTORIAL

$${}_1F_1(a; b; z) = \sum_{n=0}^{\infty} \frac{(a)_n}{(b)_n} \frac{z^n}{n!}, \quad (a)_n = a(a+1) \cdots (a+n-1)$$

EULER INTEGRAL FORM $\cdot \operatorname{Re} b > \operatorname{Re} a > 0$

$${}_1F_1(a; b; z) = \frac{1}{B(a, b-a)} \int_0^1 e^{zt} t^{a-1} (1-t)^{b-a-1} dt$$

The integral form is the one that returns: Models B2 and C2 recover P_y as exactly the Beta-weighted integral of e^{zt} .



§4.3 • TRANSFORMS

One full transform, one partial

THE BIVARIATE PGF — THE CENTRAL OBJECT

$$\begin{aligned} P(x, y) &= \mathbb{E}[x^{N_1} y^{N_2}] \\ &= \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(n_1, n_2) x^{n_1} y^{n_2} \end{aligned}$$

THE PARTIAL PGF (COHEN, 1956)

$$\begin{aligned} \tilde{P}(n_1, y) &= \mathbb{E}[y^{N_2} \mathbf{1}_{\{N_1=n_1\}}] \\ P(x, y) &= \sum_{n_1} \tilde{P}(n_1, y) x^{n_1} \end{aligned}$$

BOUNDARY FUNCTIONS (EMPTY-QUEUE TRACES)

$$P_x(x) = P(x, 0), \quad P_y(y) = P(0, y)$$

Transform only the hard direction, keep the easy class discrete: the priority class is often a plain M/M/1, so spend the analytical effort class 2 alone. This idea powers Part IV.



§3.6 • THE KEY MECHANIC

Where the derivatives come from

Jockeying and abandonment act on **every waiting customer**, so their aggregate rate is length-proportional: $\gamma_1 n_1, \theta_1 n_1$. A factor n_1 under the transform is precisely a derivative:

$$\sum_{n_1, n_2} n_1 \pi(n_1, n_2) x^{n_1} y^{n_2} = x \frac{\partial P}{\partial x}(x, y)$$

■ rate $\gamma_1 n_1$ or $\theta_1 n_1 \rightarrow$ derivative term $\rightarrow$ ODE/PDE ■ constant rate $\gamma_1 \cdot 1_{\{n_1 \geq 1\}}$ (head-of-line) $\rightarrow$ algebraic term $\gamma_1 [P - P_y]$

This single mechanic decides the difficulty of every model in the thesis.

Part II – The general model

Model X and its fundamental equation

One equation carries the whole thesis: kernel, two convection terms, two unknowns.

§4 • MODEL X – THE PARENT MODEL

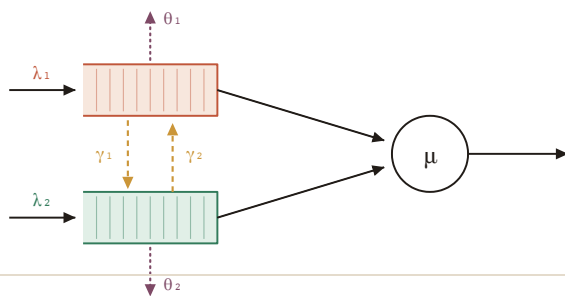
Two classes, one server, both mechanisms

Poisson arrivals λ_1, λ_2 ; common exponential service μ ; the server never interrupts a service in progress. Every **waiting** customer carries independent exponential jockeying and patience clocks.

jockeying γ_1, γ_2

abandonment θ_1, θ_2

$\rho = \lambda/\mu$



n_1, n_2 count customers **waiting** — the one in service is excluded. .
class never matters: memorylessness makes the residual service Exp either way.

§4 · STATE SPACES

One system, two coordinate systems

REGULAR SPACE S — USED THROUGHOUT

$$S = \{(0)\} \cup \{(n_1, n_2) \mid n_1, n_2 \geq 0\}$$

ALTERNATIVE SPACE $\tilde{S}$ — TRACK CLASS 2 AND THE TOTAL

$$\tilde{S} = \{(0)\} \cup \{(n_2, n) \mid n \geq n_2 \geq 0\}$$

$$n = n_1 + n_2$$

EMPTY STATES — EXACT FOR THE FULL MODEL X

$$\pi(0, 0) = \rho \pi_0, \quad \pi_0 = \frac{1}{1 + \lambda \mathbb{E}[B]}$$

A cut between idle and busy, plus renewal–reward over busy cycles: everything reduces to one scalar, the mean busy period $\mathbb{E}[B]$. Without abandonment, $\mathbb{E}[B] = (\mu - \lambda)^{-1}$ gives $\pi_0 = 1 - \rho$, $\pi(0, 0) = \rho(1 - \rho)$.

Why $\tilde{S}$? Under jockeying neither class is simple on its own — but the **total** is: jockeying conserves it. Part IV tests whether that helps.

§4.2 · BALANCE EQUATIONS ON S

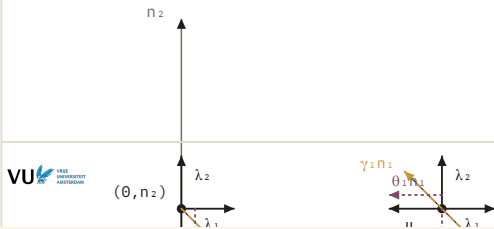
The flow at a generic state

INTERIOR STATES ($N_1 \geq 1, N_2 \geq 1$)

$$\begin{aligned} & [\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)n_1 + (\gamma_2 + \theta_2)n_2] \pi(n_1, n_2) \\ &= \mu \pi(n_1 + 1, n_2) + \lambda_1 \pi(n_1 - 1, n_2) + \lambda_2 \pi(n_1, n_2 - 1) \\ &+ \gamma_1(n_1 + 1) \pi(n_1 + 1, n_2 - 1) + \gamma_2(n_2 + 1) \pi(n_1 - 1, n_2 + 1) \\ &+ \theta_1(n_1 + 1) \pi(n_1 + 1, n_2) + \theta_2(n_2 + 1) \pi(n_1, n_2 + 1) \end{aligned}$$

Plus three boundary families ($n_1=0$; $n_2=0$; the empty state) and the idle cut $(\lambda_1 + \lambda_2)\pi_0 = \mu\pi(0, 0)$. The length-proportional rates $\gamma_i n_i$, $\theta_i n_i$

are what become derivatives.



§4.3 · LEMMA 4.2 — THE OBJECT THAT DRIVES THE THESIS

The fundamental equation of Model X

$$\begin{aligned} & [(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2] P(x, y) \\ & + xy [\gamma_1(x - y) + \theta_1(x - 1)] \frac{\partial P}{\partial x} + xy [\gamma_2(y - x) + \theta_2(y - 1)] \frac{\partial P}{\partial y} \\ & = \mu(x - y) P_y(y) - \mu x(1 - y) \pi(0, 0) \end{aligned}$$

- kernel — exactly Model A's algebraic coefficient
- jockeying factors vanish on the diagonal $x=y$
- abandonment factors vanish at $x=1$ (resp. $y=1$)
- the two unknowns



§4.3 · REMARK 4.3

The vanishing factors choose the technique

- Model A · no convection

algebraic kernel

No derivative terms. The kernel vanishes on a curve; evaluate at the root $\mathbf{x}^*(\mathbf{y})$ inside the unit disc.

- Model B₂ · γ_i only

$(x-y) \rightarrow$ interior singularity

The singularity sits at the diagonal point $\mathbf{x}=\mathbf{y}$, with negative exponent **analyticity** there pins $P_y(y)$.

- Model C₂ · θ_i only

$(x-1) \rightarrow$ boundary singularity

The singularity moves to $\mathbf{x}=1$ on the unit circle, with positive exponent: mere **boundedness** suffices.

- Model B[#]

- Model C[#]

head-of-line $\rightarrow$ collapse

Constant rate $1\{n_i \geq 1\}$ removes the derivative outright — back to Model A's algebraic form.

§4 · DEDICATED RESULT — MODEL X

Theorem 4.1 — two identities pin the empty states

THEOREM 4.1 empty-state probabilities of the general model

Before any model is solved: for the full Model X, with both mechanisms active,

$$\pi(0, 0) = \rho \pi_0$$

$$\pi_0 = \frac{1}{1 + \lambda \mathbb{E}[B]}, \quad \pi(0, 0) = \frac{\rho}{1 + \lambda \mathbb{E}[B]}$$

Proof. flux cut $\lambda \pi_0 = \mu \pi(0, 0)$ between the idle and busy states · renewal–reward over regeneration cycles at the empty state — exact for every (γ, θ) . Everything reduces to one scalar, $\mathbb{E}[B]$.

MODEL PRIMITIVES

λ_1, λ_2	Poisson arrival rates — class 1 has non-preemptive priority over class 2
μ	exponential service rate, common to both class
λ, ρ	$\lambda = \lambda_1 + \lambda_2$ total arrival rate; $\rho = \lambda/\mu$ the offered load
γ_i, θ_i	jockeying / abandonment rates — arbitrary ≥ 0 the theorem holds for the whole family

OBJECTS IN THE STATEMENT

π_0	stationary probability of the empty system — server idle, nobody waiting
$\pi(0, 0)$	server busy , both queues empty: $N_1 = N_2 = 0$ waiting
$\mathbb{E}[B]$	mean busy period : from an arrival into an empty system until the system next empties
assumption	the CTMC of Model X is positive recurrent, so th stationary distribution exists

• **Corollary 4.4** No abandonment $\Rightarrow$ the M/M/1 baseline, for A, B, B₂ and B^H at once

EXP

Part III — Analysis

The solved models — and the one that is not

Baseline A, the open Model B, one-way jockeying B₂,
abandonment C₂, and the head-of-line pair.

Seven models, one functional equation

MODEL	ACTIVE MECHANISM	EQUATION TYPE	STATUS
X	jockeying & abandonment	PDE · ???	• Model X
A	none — baseline	algebraic PGF	• Model A
B	bidirectional jockeying	PDE · Volterra family	• Model B
B ₂	one-way jockeying (1→2), $\gamma_1 n_1$	ODE in x · Kummer	• Model B ₂
C ₂	class-1 abandonment, $\theta_1 n_1$	ODE in x · Kummer	• Model C ₂
B ^H	jockeying, head-of-line $\gamma_1 \cdot 1\{n_1 \geq 1\}$	algebraic PGF	• Model B ^H
C ^H	abandonment, head-of-line $\theta_1 \cdot 1\{n_1 \geq 1\}$	algebraic PGF	• Model C ^H



Three complementary tools

Kernel method

The algebraic coefficient of P vanishes on a curve. Evaluating there forces the RHS to vanish and pins the unknowns.

Models A, B^H, C^H.

Integrating factor

A surviving $\partial/\partial x$ term gives a first-order linear ODE in x ; a regularity demand at its singularity fixes the constant.

Models B₂ (analyticity), C₂ (boundedness).

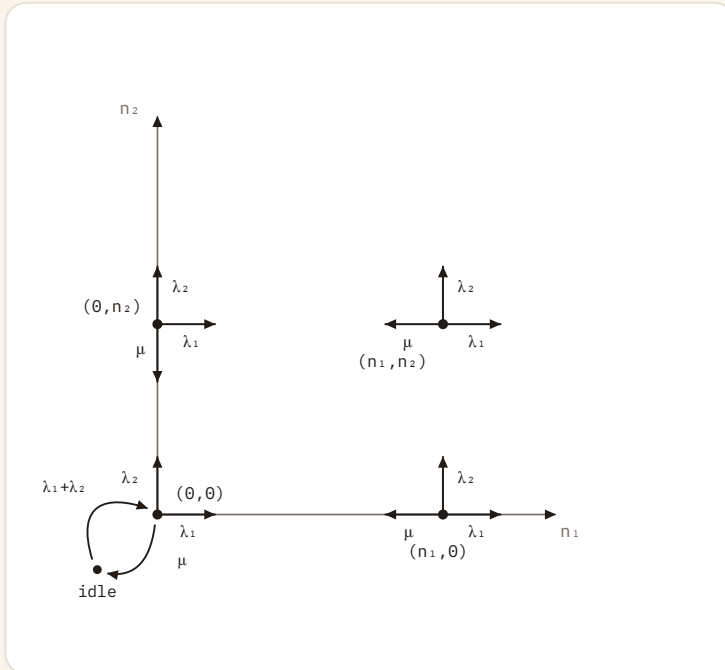
Pollaczek–Khinchine

Class 2 is an M/G/1 queue whose service time is a class-1 **busy period** — an independent confirmation.

Models A, C₂. Jockeying destroys it.



No mechanisms: the equation is algebraic



REDUCED FUNDAMENTAL EQUATION

$$[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2] P(x, y) = \mu(x - y)P_y(y) - \mu x(1 - y)\pi(0, 0)$$

Fix y and find the root $x^*(y)$ of the kernel inside the unit disc. bounded there, so the LHS vanishes — hence the RHS must t

$$P_y(y) = \frac{x^*(y)(1 - y)}{x^*(y) - y} \pi(0, 0)$$

THE EMPTY STATE — FOR COMPLETENESS

$$\pi(0, 0) = (1 - \rho) \rho$$



§5 • DEDICATED RESULT — MODEL A

Theorem 5.1 — the baseline PGF

THEOREM 5.1 closed-form PGF of Model A

Under $\rho < 1$, the joint PGF is

$$P(x, y) = \frac{\rho(1 - \rho) y(1 - y)}{(1 + \rho_1 + \rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2} \cdot \frac{x - x^*(y)}{x^*(y) - y}$$

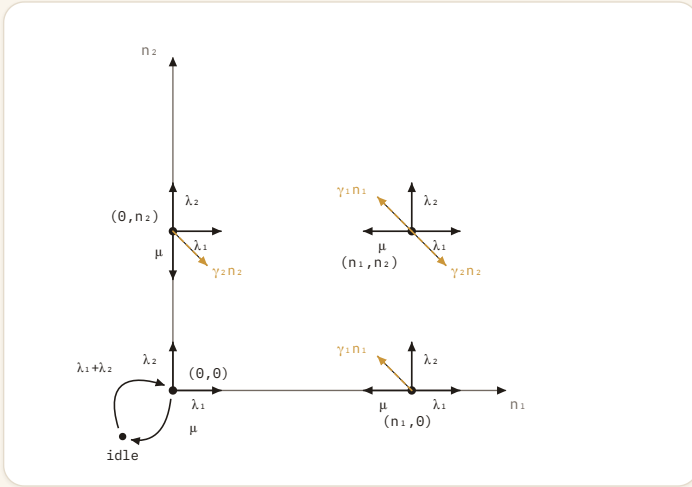
$$x^*(y) = \frac{[\mu + \lambda_1 + \lambda_2(1 - y)] - \sqrt{[\mu + \lambda_1 + \lambda_2(1 - y)]^2 - 4\lambda_1\mu}}{2\lambda_1}$$

Proved twice. kernel method (analytic) · Poisson splitting + PASTA (probabilistic) — the two routes meet in the same root $x^*(y)$. Classical result, recovered as calibration for everything that follows.



§6 • MODEL B — BIDIRECTIONAL JOCKEYING ($\Gamma_1, \Gamma_2 > 0$)

Now both derivatives survive



MODEL-B FUNDAMENTAL EQUATION

$$\begin{aligned} & [(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2] P \\ & + \gamma_1 x y (x - y) \frac{\partial P}{\partial x} + \gamma_2 x y (y - x) \frac{\partial P}{\partial y} \\ & = \mu(x - y) P_y(y) - \mu x (1 - y) \pi(0, 0) \end{aligned}$$

Both convection terms carry $(x-y)$: the Method of Characteristics applies, along the straight lines

$$C_1 = \gamma_2 x + \gamma_1 y$$

And jockeying destroys the Poisson split of arrivals — the probabilistic route is gone too.

§6 · THE FIRST OBSTRUCTION

Model B reduces to a Volterra-type integral equation

CLAIM 6.3 integral representation along characteristics

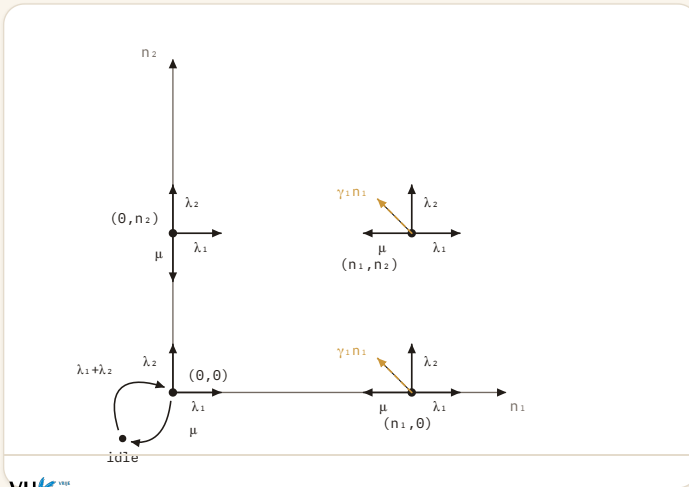
Along each characteristic line, variation of parameters gives $P = P_h \cdot \int h/P_h$. Evaluating on the diagonal $x=y=z$ — where $P(z,z)=\pi(0,0)/(1-\rho z)$ is already known — the boundary function must satisfy a **Volterra integral equation of the first kind**:

$$\mathcal{F}(z) = \int_0^z \mathcal{K}(z, t) P_y(y(t; z)) dt$$

The forcing depends on the unknown boundary itself. Inverting it needs tools beyond this thesis — **Model B is left open**, and it is the representative of every model with a $\partial/\partial y$ term, Model X included.

§7 · MODEL B₂ — ONE-WAY JOCKEYING ($\Gamma_1 > 0, \Gamma_2 = 0$)

Kill the $\partial/\partial y$ term, keep the jockeying



REDUCED EQUATION — AN ODE IN X FOR EACH FIXED Y

$$[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2]P + \gamma_1 x y (x - y) \frac{\partial P}{\partial y} = \mu(x - y)P_y(y) - \mu x(1 - y)\pi(0, 0)$$

The integrating factor has a singularity at the **interior** point with negative exponent $\beta(y)$. A PGF must be analytic there — that single demand determines $P_y(y)$.

$$\alpha(y) = \mu / (\gamma_1 y) \quad \beta(y) = (1 - y)(\lambda y - \mu) / (\gamma_1 y)$$

§7 • DEDICATED RESULT — MODEL B₂

Theorem 7.1 — the B₂ generating function

THEOREM 7.1 boundary function and PGF of Model B₂

Under $\rho < 1$, analyticity of P at the interior point $x = y$ forces

$$P_y(y) = \frac{\mu \pi(0, 0)}{\mu - \lambda y} \cdot \frac{{}_1F_1(\alpha(y) + 1; b^*(y); -\lambda_1 y / \gamma_1)}{{}_1F_1(\alpha(y); b^*(y); -\lambda_1 y / \gamma_1)}$$

$$P(x, y) = \frac{\mu e^{\lambda_1 x / \gamma_1}}{\gamma_1 y x^{\alpha(y)} (y - x)^{\beta(y)}} \left[P_y(y) \mathcal{H}_1(x, y) + (1 - y) \pi(0, 0) \mathcal{H}_2(x, y) \right]$$

valid on $0 \leq x \leq y$; extends by continuation across $x = y$.

THE HELPER INTEGRALS

$$\mathcal{H}_1(x, y) = \int_0^x e^{-\lambda_1 t / \gamma_1} t^{\alpha(y)-1} (y - t)^{\beta(y)} dt$$

$$\mathcal{H}_2(x, y) = \int_0^x e^{-\lambda_1 t / \gamma_1} t^{\alpha(y)} (y - t)^{\beta(y)-1} dt$$

$\mathcal{H}_1$ weighs $P_y(y)$; $\mathcal{H}_2$ weighs $(1 - y)\pi(0, 0)$ — exponents traded between $t = 0$ and $t = y$.

§8 • MODEL C₂ — CLASS-1 ABANDONMENT ($\theta_1 > 0$)

Abandonment moves the singularity to $x = 1$

REDUCED EQUATION

$$\begin{aligned} & [\text{kernel} + \gamma_1 y(x - y)] P \\ &= [\mu(x - y) + \gamma_1 y(x - y)] P_y - \mu x(1 - y) \pi(0, 0) \end{aligned}$$

$$\begin{aligned} & [\text{kernel} + \theta_1 y(x - 1)] P \\ &= [\mu(x - y) + \theta_1 y(x - 1)] P_y - \mu x(1 - y) \pi(0, 0) \end{aligned}$$

Both are solved by the Kernel Method exactly as Model A. For C^H even stability relaxes: $(\mu + \theta_1)(\mu - \lambda) + \lambda_1 \theta_1 > 0$ — weaker than $\rho < 1$.



§9 · DEDICATED RESULT — MODEL B^u

Theorem 9.1 — head-of-line jockeying is rational

THEOREM 9.1 closed-form PGF of Model B^u

Under $\rho < 1$, the joint PGF is the rational function

$$P(x, y) = \frac{\mu \rho (1 - \rho) (1 - y)}{\lambda_1 (x^*(y) - y) (x^+(y) - x)}$$

$$\lambda_1 x^2 - [\lambda_1 + \lambda_2(1 - y) + \mu + \gamma_1] x + (\mu + \gamma_1 y) = 0$$

Proof. the constant rate $\gamma_1 \cdot \mathbf{1}\{n_1 \geq 1\}$ joins the kernel as an algebraic term — no derivative survives, and the Kernel Method runs exactly as for Model A. CTMC check: max error $< 3 \times 10^{-7}$ on a 5×5 grid. As $\gamma_1 \rightarrow 0^+$ the formula collapses to Theorem 5.1.

EVERY SYMBOL IN THE STATEMENT

γ_1	head-of-line jockeying: only the first waiting class-1 customer may switch, so the aggregate rate is the constant $\gamma_1 \cdot \mathbf{1}\{n_1 \geq 1\}$ ($\gamma_2 = \theta_1 = \theta_2 = 0$)
λ, ρ	$\lambda = \lambda_1 + \lambda_2$, $\rho = \lambda / \mu < 1$ — stability; the total count still conserved
$P(x, y)$	joint PGF $E[x^{N_1} y^{N_2}]$ of the waiting counts, ov busy states
$\rho(1 - \rho)$	the numerator constant = $\pi(0, 0)$, by Corollary 9 below
$x^*(y), x^+(y)$	the two roots of the kernel quadratic on the left x^* inside the unit disc (where boundedness pir the RHS), x^+ outside
$\mu + \gamma_1 y$	the constant term of the quadratic: service plus the jockey stream re-entering through Queue 2



§10 · DEDICATED RESULT — MODEL C^u

Theorem 10.1 — head-of-line abandonment stays rational

THEOREM 10.1 closed-form PGF of Model C^u

Under $(\mu + \theta_1)(\mu - \lambda) + \lambda_1 \theta_1 > 0$, the joint PGF is the rational function

$$P(x, y) = \frac{\mu(\mu + \theta_1) (1 - y) \pi(0, 0)}{\lambda_1 (x^+(y) - x) [(\mu + \theta_1 y) x^*(y) - y(\mu + \theta_1)]}$$

$$\lambda_1 x^2 - [\lambda_1 + \lambda_2(1 - y) + \mu + \theta_1] x + (\mu + \theta_1) = 0$$

Proof. Kernel Method again — abandonment enters the kernel as the algebraic term $\theta_1 y(x - 1)$. CTMC check: max error $< 3 \times 10^{-11}$ on a 5×5 grid. As $\theta_1 \rightarrow 0^+$ the formula recovers Theorem 5.1 exactly.

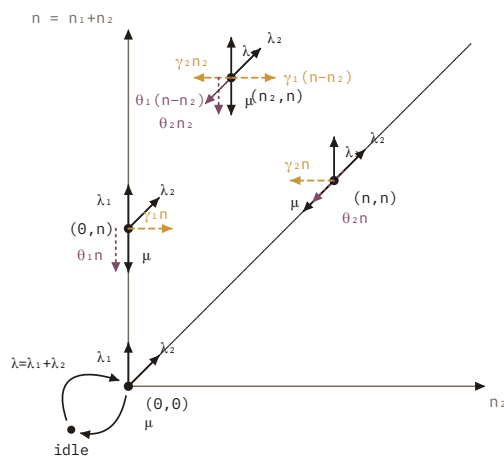
Part IV – A documented negative result

The $\tilde{S}$ experiment

If jockeying conserves the total, why not make the total a coordinate? Because the recurrence refuses to close.

§4.2 · BALANCE EQUATIONS ON S'

Model X in the (n_2, n) coordinates



INTERIOR STATES ($1 \leq n_2 \leq N-1, N \geq 2$)

$$\begin{aligned} & [\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)(n - n_2) + (\gamma_2 + \theta_2)n_2] \tilde{\pi}(n_2, n) \\ &= \lambda_1 \tilde{\pi}(n_2, n-1) + \lambda_2 \tilde{\pi}(n_2-1, n-1) + \mu \tilde{\pi}(n_2, n+1) \\ & \quad + \gamma_1(n - n_2 + 1) \tilde{\pi}(n_2-1, n) + \gamma_2(n_2 + 1) \tilde{\pi}(n_2 + 1, n) \\ & \quad + \theta_1(n + 1 - n_2) \tilde{\pi}(n_2, n+1) + \theta_2(n_2 + 1) \tilde{\pi}(n_2 + 1, n+1) \end{aligned}$$

Arrivals now move diagonally (λ_2 raises both n_2 and n); jockeying moves horizontally at fixed n ; the diagonal $n_2 = n$ means **all waiters are class 2** — it will matter.

§4.3 • LEMMA 4.5 — THE PPGF EQUATION

The fundamental equation on $\tilde{S}$

Transform only n_2 and keep the total n discrete: $\tilde{P}(y, n) = \sum \tilde{\pi}(n_2, n) y^{n_2}$, a polynomial of degree n .

$$\begin{aligned} & [-(\gamma_2 + \gamma_1 y)(1 - y) + (\theta_2 - \theta_1)y] \frac{d}{dy} \tilde{P}(y, n) + [\lambda_1 + \lambda_2 + \mu + \gamma_1 n(1 - y) + \theta_1 n] \tilde{P}(y, n) \\ &= (\lambda_1 + \lambda_2 y) \tilde{P}(y, n-1) + [\mu + \theta_1(n+1)] \tilde{P}(y, n+1) + (\theta_2 - \theta_1 y) \frac{d}{dy} \tilde{P}(y, n+1) \\ & \quad + \mu y^n (1 - y) \tilde{\pi}(n+1, n+1) \end{aligned}$$

A differential-**difference** equation: levels $n-1, n, n+1$ are coupled — plus **one forcing term**.

§5.3 • REMARK 4.6 + APPROXIMATION 5.7

Why the experiment did not work

MODEL A ONLY: DROP THE FORCING (IT DECAYS LIKE P^n)

$$\tilde{P}(y, n) \approx \rho(1 - \rho) [\tilde{y}^*(y)]^n, \quad \tilde{y}^*(y) = \frac{(\lambda + \mu) - \sqrt{(\lambda + \mu)^2 - 4\mu(\lambda_1 + \lambda_2 y)}}{2\mu}$$

The forcing term $\mu y^n (1 - y) \tilde{\pi}(n+1, n+1)$ feeds in the **diagonal** states — all waiting customers class 2. Those probabilities are exactly what we are solving for: the recurrence **does not close on itself**.

Instead of collapsing 2-D balance to a 1-D recurrence, the change of coordinates makes things **harder** — for every model with an active mechanism, $\tilde{S}$ is strictly worse than !

Even then it is an **approximation, not a theorem** — accurate only where the forcing decays faster than the solution, $\rho < \tilde{y}^*(y)$: a priority-head moderate load. At $y=1$ it does recover the M/M/1 marginal $(1-\rho)\rho^{n+1}$.

deliberate negative result

Part V – Results

Validate, then compare

Every closed form against an exact CTMC; then jockeying and abandonment head to head.

§11 · VALIDATION

Every closed form matches the CTMC

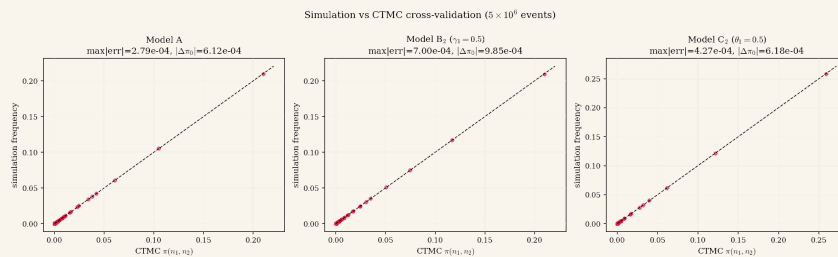


Fig. 1. Analytic PGF marginals against a truncated CTMC that recovers the baseline as the mechanism rates vanish.

simulation vs. CTMC · all solved models

CONSERVATION CHECK (C_2)

$<10^{-14}$

offered = carried + lost

BASELINE RECOVERY

exact

mechanism rates $\rightarrow 0 \Rightarrow$ Model A

Jockeying redistributes, it does not shorten

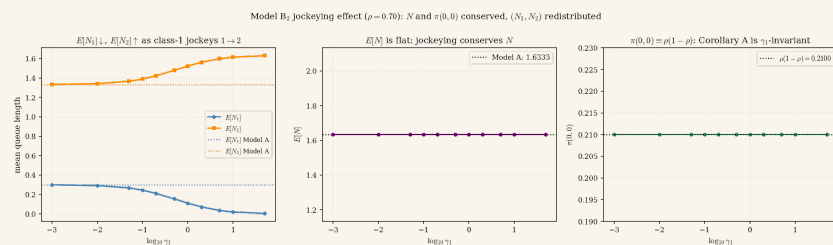


Fig. 2. As γ_1 grows, $E[N_1]$ falls and $E[N_2]$ rises — the total $E[N]$ and $\pi(0,0)$ stay pinned to the baseline.

Model B₂ • $\rho = 0.70$ • truncated CTMC

MEAN QUEUE $E[N_1]$

0.155 ↓

vs. $0.30 \cdot$ Model A

TOTAL $E[N]$

1.633 →

conserved — identical to A

WAITING RATIO W_1/W_2

0.139 ↓

vs. $0.30 \cdot$ Model A

Abandonment is the mechanism that shortens

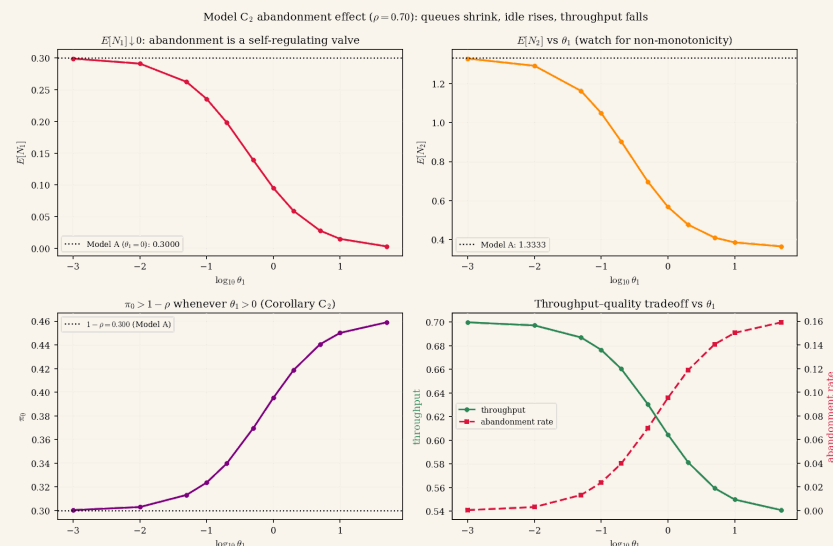


Fig. 3. As θ_1 grows, work leaves the system: $E[N]$ drops well below the baseline and class-1 loss rises.

Model C₂ • $\rho = 0.70$ • truncated CTMC

TOTAL $E[N]$

0.836 ↓

vs. $1.633 \cdot$ Model A

CLASS-1 LOSS FRACTION

23.2 %

$\theta_1 = 0.5$

MEAN QUEUE $E[N_1]$

0.139 ↓

vs. $0.30 \cdot$ Model A

The abandonment gap widens with load

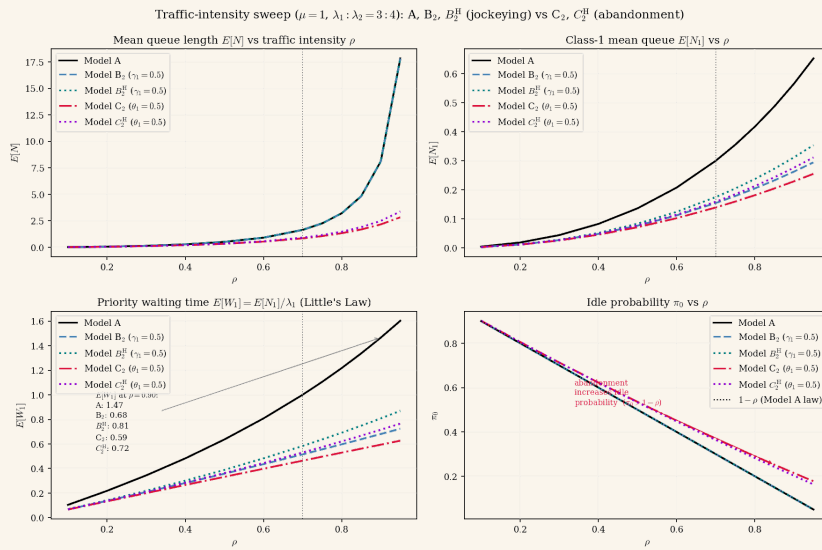


Fig. 4. A, B₂ and B^H coincide in $E[N]$ and obey the $1-\rho$ idle law — jockeying conserves N. C₂ and C^H sit below in $E[N]$ and above in π_0 at every load; the head-of-line pair interpolates.

five solved models · $\mu = 1 \cdot \lambda_1 : \lambda_2 = 3 : 4 \cdot \gamma_1 = \theta_1 = 0.5$

CLASS-2 QUEUE $E[N_2] \cdot P = 0.90$

1.925 ↓

vs. 7.535 · Model A

IDLE PROBABILITY $\pi_0 \cdot P = 0.70$

0.370 ↑

vs. 0.300 = $1-\rho$ · A and B₂

CLASS-2 WAIT $E[W_2] \cdot B_2$

+10.9% ↑

3.697 vs. 3.333 · jockeying penalises class 2



§11 · STRUCTURAL COMPARISON — TABLE 11.1

Five solved models, side by side

	A	B ₂	B ^H	C ₂	C ^H
MECHANISM RATE	none	$\gamma_1 n_1$	$\gamma_1 \cdot 1\{n_1 \geq 1\}$	$\theta_1 n_1$	$\theta_1 \cdot 1\{n_1 \geq 1\}$
FUNDAMENTAL EQ.	algebraic	1st-order ODE	algebraic	1st-order ODE	algebraic
PINNING OF P _Y	kernel root x^*	analyticity at $x=y$	kernel root x^*	boundedness at $x=1$	kernel root x^*
P(X,Y) FORM	rational	integral + $\gamma_1 F_1$	rational	integral + $\gamma_1 F_1$	rational
CONSERVES N	yes	yes	yes	no	no
$\pi(0,0)$	$\varrho(1-\varrho)$	$\varrho(1-\varrho)$	$\varrho(1-\varrho)$	Cor. 8.2	Cor. 10.2

Head-of-line rates collapse the ODE back to the algebraic form — $P(x,y)$ becomes rational again.



§11.4 · CONVERGENCE TO THE BASELINE

Linear convergence — along two different paths

As its mechanism rate $\rightarrow 0^+$, each specialisation must recover Model A. The L1 distance $\|\pi - \pi_A\|_1$ falls **first-**

Jockeying: interior redistribution

order: log–log slopes of 0.85–0.93 against the theoretical 1, the shortfall explained by CTMC truncation.

B_2 · slope 0.87

B'' · slope 0.93

C_2 · slope 0.85

C'' · slope 0.90

π_0 and $\pi(0,0)$ stay pinned at $1-p$ and $p(1-p)$ for **every** γ , — convergence is invisible at the empty states and happens entirely in the interior.

Abandonment: drift and relax

A true departure moves the idle descriptors above the baseline; then relax back onto it only as $\theta_1 \rightarrow 0^+$. The conservation dichotomy, one last time.

§12 · CONCLUSION

Four mechanisms solve, two obstruct

- 1 **Kernel root** inside the disc (A) · **analyticity** at $x=y$ (B_2) · **boundedness** at $x=1$ (C_2) · **head-of-line collapse** (B'' , C'') — the four ways $P_y(y)$ gets pinned.
- 2 The obstructions: a **$\partial/\partial y$ term** forces a first-kind Volterra equation (B); Model X adds transcendental characteristics on top.
- 3 **Jockeying redistributes** delay at fixed total work; **abandonment reduces** the load itself. Same waiting room, opposite registers.

§12 · FUTURE WORK

What the machinery makes possible next

Head-of-line ladder

Invert the Volterra equation

Let the first k customers act: $\min(n_1, k)$ keeps the equation algebraic at the price of k boundary unknowns; expect convergence to B_2 / C_2 as $k \rightarrow \infty$.

Exact means from the PGFs

Differentiate at $(1,1)$ — with L'Hôpital at the $x=1$ singularity — to replace the numerical $E[N_i]$, $E[W_i]$ with closed forms.

Model B's kernel and forcing are explicit — numerical inversion (or sharper analysis) would close the first obstruction.

The clinically wanted direction

The $2 \rightarrow 1$ upgrade of a deteriorating patient is exactly $\gamma_2 > 0$ — the Volterra-hard direction. Calibration to elderly-care data comes after



THANK YOU

Questions welcome.

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